# Revisiting fine-root traits in the global spectrum of plant form and function

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For decades, plant ecology has searched for a single organizing principle of plant strategy, often framed as a global spectrum of plant form and function (GSPFF). Carmona et al. [2021] put this idea to the test by combining global aboveground traits with fine-root traits in a unified ordination. In detail, Carmona et al. [2021] integrate the TRY and GRooT datasets, yielding 301 species with complete aboveground and fine-root trait information. To increase sample size, missing values were then imputed using missForest, expanding the dataset to 1,218 species. A principal component analysis (PCA) with varimax rotation was applied to examine the trait structure. Their conclusion is striking: plant strategies do not collapse onto one dominant axis. Instead, trait variation unfolds in a four-dimensional space that separates into two largely independent planes, an aboveground and a fine-root spectrum. The aboveground space itself shows clear clustering by growth form, with herbaceous species and angiosperm trees occupying distinct regions. Accordingly, species with similar root characteristics can differ strongly in their aboveground traits, suggesting that what appears coordinated at first glance reveals a more fragmented picture when examined more closely.

If robust, this result challenges the notion of a coherent whole-plant economics spectrum and has far-reaching implications for trait-based prediction and ecosystem modeling. We revisit this conclusion with an independent robustness analysis of the ordination geometry. Specifically, we test whether the reported decoupling persists under alternative rotation choices, different imputation schemes, explicit uncertainty propagation, and ecological subsetting.

The reported aboveground–fine-root decoupling is inferred from a trait space that is subsequently populated using imputed values for most species. To ensure that this key geometric pattern is not driven by missing-data handling, we first test the sensitivity of the PCA configuration to the imputation method and to uncertainty in imputed values. To assess the sensitivity of the functional trait space to missing-data treatment, we first compared the original missForest imputation with a deliberately naive mean-imputation. Despite ignoring phylogenetic and multivariate structure, mean imputation produced nearly identical principal component axes and species configurations. A Procrustes comparison between the two four-dimensional configurations revealed extremely high correspondence (ProcCorr = 0.979; $p = 0.0001$), indicating that the large-scale geometry is data-driven rather than imputation-driven (Fig. 1).

In addition, testing the effect of varimax rotation confirmed that rotation does not induce artificial separation between domains: correspondence between rotated aboveground and fine-root domains and their representation in the combined space remained very high (ProcCorr = 0.988 / 0.982; $p = 0.0001$), whereas cross-domain correspondence remained low (ProcCorr = 0.178; $p = 0.0001$). While missForest may improve individual predictions, it offers little advantage for reconstructing the global trait space.

We then propagated missForest out-of-bag (OOB) errors through the PCA using a parametric bootstrap, adding trait-specific Gaussian noise to all originally imputed values. Across 50 bootstrap replicates, Procrustes residual mismatch relative to the reference configuration remained low (mean $m^2 = 0.058$; $n = 1218$), even under a conservative scenario in which OOB-derived uncertainty was doubled (mean $m^2 = 0.058$; $n = 1218$). In both cases, deviations were small relative to total ordination spread and did not alter axis interpretation. Thus, neither the choice of imputation method nor realistic or inflated imputation uncertainty meaningfully reshapes the global trait space.

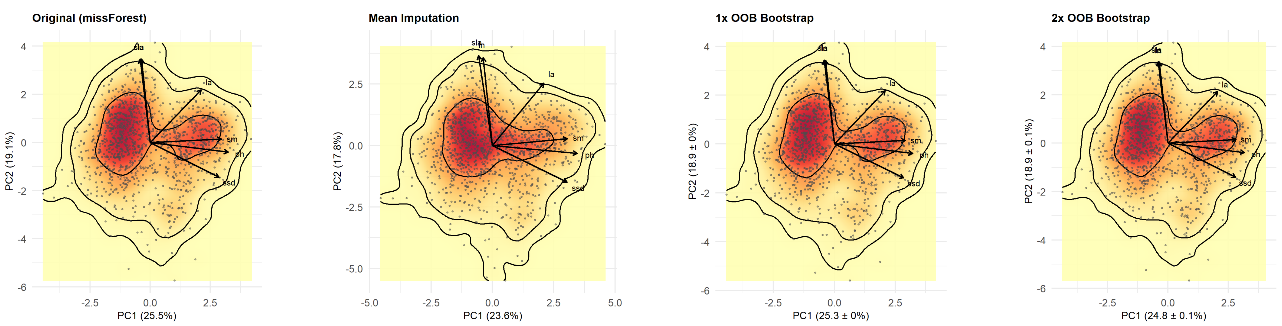


**Figure 1:** PCA of aboveground traits (PC1–PC2) under four imputation scenarios: original missForest impuation, naive mean imputation, OOB-based bootstrap (1×), and inflated OOB bootstrap (2×).

Finally, we asked whether the decoupled structure of the trait space is globally consistent or shaped by particular ecological contexts. For each major biome (temperate, tropical, continental), we refitted biome-specific PCAs.

Biome-specific ordinations are shown in Figure 2. In some subsets, fewer effective dimensions were retained, indicating that trait variation may concentrate into fewer dominant gradients within biomes compared to the global space. In contrast to the global configuration, the tropical subset does not display the pronounced two-cluster pattern in the aboveground plane (PC1–PC2) observed in the full dataset. Instead, species occupy a narrower region of the trait space, indicating reduced variation within this biome. This pattern is consistent with the lower functional dissimilarities across biomes reported by Carmona et al. [2021].

To formally assess geometric stability, we compared biome-specific configurations to the global reference using Procrustes analysis across the first three principal components (PC1–PC3). Temperate assemblages showed near-perfect correspondence (ProcCorr $= 0.99$; $p < 0.001$; $n = 71$), as did continental assemblages (ProcCorr $= 0.98$; $p < 0.001$; $n = 40$), indicating that these biomes represent well-sampled subsets of the broader gradients. Tropical assemblages exhibited somewhat lower but still substantial alignment (ProcCorr $= 0.88$; $p < 0.001$; $n = 27$).

Overall, although biome-specific PCAs may show reduced variance and weaker visual separation between above- and fine-root traits, the overall geometric structure remains aligned with the global configuration. Thus, the observed compression likely reflects restricted trait variance and limited sample size rather than a fundamental restructuring of trait relationships. Nevertheless, given the comparatively small sample sizes of some subsets—particularly the tropical biome—these results should be interpreted cautiously, and re-evaluation with larger regional datasets would strengthen conclusions about biome-level generality.

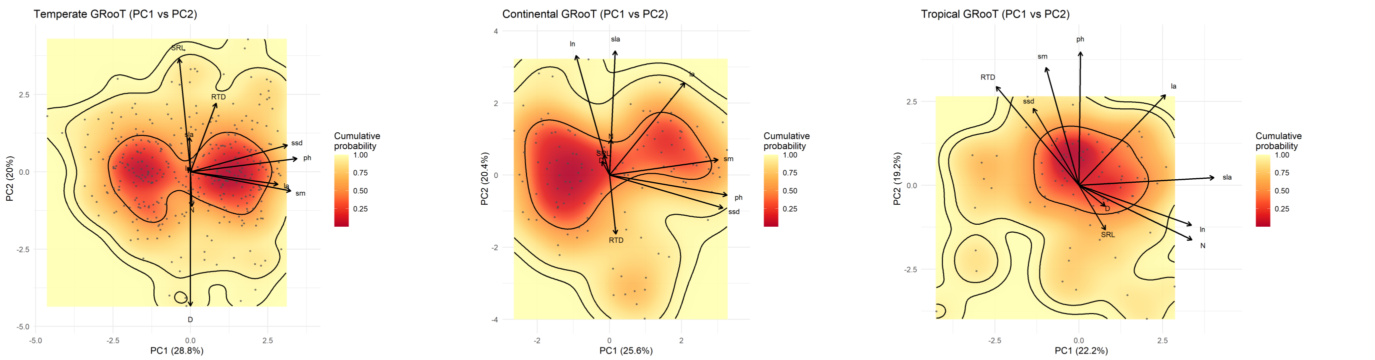


**Figure 2:** Functional space (PC1–PC2) for continental (A), temperate (B) and tropical (C) biome according to the GRooT biome separation.

Our reanalysis demonstrates that the reported aboveground–fine-root decoupling is not a rotation artefact, nor an artefact of imputation or uncertainty inflation, and remains stable across ecological subsets. This structural robustness implies that fine-root traits genuinely expand the dimensionality of the global plant spectrum, rather than simply mirroring aboveground axes. The evidence does not support a single whole-plant economics axis; instead, fine-root traits contribute independent dimensions to plant strategy space. However, it is important to clarify that belowground here refers specifically to fine-root traits, which primarily represent acquisitive functions such as nutrient and water uptake—functionally analogous to leaves rather than to the entire aboveground plant. In contrast, the aboveground trait set spans size, tissue economics, and reproductive strategy, leading to an inherent asymmetry between compartments. Additional root system traits, such as those related to storage or structural support, may further expand belowground dimensionality. Thus, the observed dimensional asymmetry should not be over-generalized. Future work should extend this framework to include a broader range of belowground traits to fully capture whole-plant diversity.

## References

Carlos P. Carmona, C. Guillermo Bueno, Aurele Toussaint, Sabrina Träger, Sandra Díaz, Mari Moora, Alison D. Munson, Meelis Pärtel, Martin Zobel, and Riin Tamme. Fine-root traits in the global spectrum of plant form and function. *Nature*, 597(7878):683–687, September 2021. ISSN 0028-0836, 1476-4687. doi: 10.1038/s41586-021-03871-y.